

SOVABIDS: EEG-to-BIDS conversion software focused on automation, reproducibility and interoperability

Yorguin-José Mantilla-Ramos^{1,5,6,7,8,9¶}, Brayan-Andrés Hoyos-Madera^{1,5},
Steffen Bollmann², Aswin Narayanan², David White⁴, Oren
Civier^{3,4}, and Tom Johnstone^{3,4}

¹ Grupo Neuropsicología y Conducta (GRUNECO), Universidad de Antioquia, Medellín, Colombia ² The University of Queensland, Brisbane, Queensland, Australia ³ Australian National Imaging Facility, Australia ⁴ Swinburne University of Technology, Melbourne, Victoria, Australia ⁵ Semillero de Investigación Neurociencias Computacionales (NeuroCo), Universidad de Antioquia, Medellín, Colombia ⁶ Cognitive and Computational Neuroscience Laboratory (CoCo Lab), Psychology Department, Université de Montréal, Montréal, Canada ⁷ Mila (Quebec AI Institute), Montréal, Canada ⁸ Grupo Sistemas Embebidos e Inteligencia Computacional (SISTEMIC), Facultad de Ingeniería, Universidad de Antioquia, Medellín, Colombia ⁹ Semillero de Investigación Machine Learning and Robotics, Facultad de Ingeniería, Universidad de Antioquia, Medellín, Colombia ¶ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 06 August 2025

Published: unpublished

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Summary

Electroencephalography (EEG) data are used in many fields, from neuroscience to clinical research, but recordings are stored in a wide variety of formats and lab-specific folder structures, making them difficult to organize, share, or compare across studies. SOVABIDS is an open-source tool that helps researchers convert EEG data into the Brain Imaging Data Structure (BIDS) ([Gorgolewski et al., 2016](#)), a standard that promotes FAIR data practices (Findability, Accessibility, Interoperability, and Reusability) and is increasingly being adopted for data sharing and large-scale collaboration. Rather than manually renaming files or writing dataset-specific scripts, SOVABIDS allows users to define conversion rules in human-readable YAML configuration files, which are then applied automatically across all participants in a dataset ([Figure 1](#) illustrates this process). The tool is usable as a Python package, command-line tool, or interactive terminal user interface (TUI), and its API supports integration with graphical frontends for users who prefer a visual interface. Comprehensive documentation and tutorials are available at sovabids.readthedocs.io.

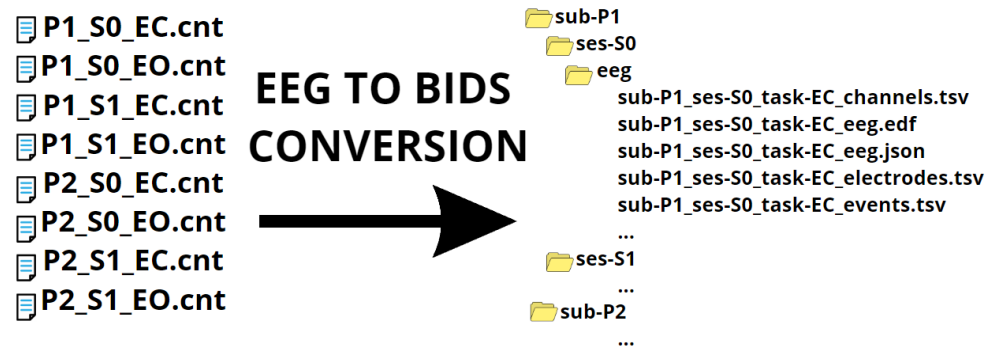


Figure 1: Illustration of the EEG to BIDS conversion. The left side shows raw EEG files with participant-specific naming conventions (for example, P1_S0_EC.cnt), where P1 and P2 represent participants, S0 and S1 indicate sessions, and EC (Eyes Closed) and EO (Eyes Open) refer to tasks. These raw files are converted into the BIDS format, shown on the right, where data are systematically organized into subject (sub-), session (ses-), and modality (eeg) folders. Each EEG recording is saved in standardized BIDS-compliant formats, including .edf for EEG signals and .tsv/.json for metadata.

29 Statement of need

30 Electroencephalography (EEG) is a widely used neuroimaging technique that provides
31 high temporal resolution for studying brain activity. Its applications span numerous
32 fields, including cognitive neuroscience, clinical diagnostics, brain-computer interfaces,
33 and neuroengineering. With the increasing volume and complexity of EEG data, ensuring
34 reproducibility, standardization, and interoperability has become a growing priority in the
35 field. The Brain Imaging Data Structure extension for EEG (EEG-BIDS) (Pernet et al.,
36 2019) provides a consistent framework for organizing EEG datasets, facilitating data sharing
37 (Markiewicz et al., 2021), large-scale collaborations, cross-study comparisons, and promoting
38 FAIR data practices (Wilkinson et al., 2016) across a wide range of research applications from
39 fundamental cognitive neuroscience to large-scale clinical neuroimaging.

40 Despite these advantages, converting EEG datasets to BIDS in practice remains a significant
41 challenge. EEG data varies widely in format due to the multitude of proprietary standards used
42 by different hardware vendors, and datasets are often organized according to lab-specific or
43 acquisition-driven conventions rather than any consistent structure. For researchers without
44 strong programming backgrounds, particularly those at smaller or less well-resourced institutions,
45 this conversion process typically requires either substantial manual effort or the ability to write
46 dataset-specific scripts, both of which are error-prone and difficult to reproduce or scale to
47 large multi-participant studies. SOVABIDS addresses this gap by providing a rule-based,
48 semi-automated conversion workflow that is accessible to users with limited programming
49 experience while remaining flexible enough to handle the heterogeneity common in real-world
50 EEG datasets.

51 State of the Field

52 Several tools support EEG conversion to BIDS, including MNE-BIDS (Appelhoff et al., 2019),
53 data2bids in FieldTrip (Oostenveld et al., 2011), EEG-BIDS in EEGLAB (Delorme & Makeig,
54 2004), and EEG2BIDS (Rogers et al., 2022). More general-purpose converters such as Bidsme
55 (Beliy et al., 2023) also support EEG alongside other modalities.

56 MNE-BIDS (Appelhoff et al., 2019) provides a powerful programmatic interface within the
57 MNE ecosystem and offers fine-grained control over metadata specification. However, it

typically requires dataset-specific scripting, which limits accessibility for non-technical users and reduces scalability for large multi-participant studies.

FieldTrip (Oostenveld et al., 2011) and EEGLAB (Delorme & Makeig, 2004) provide conversion utilities integrated within their respective analysis environments. While convenient for users already embedded in those ecosystems, conversion workflows often require manual interaction or scripting for each dataset.

EEG2BIDS (Rogers et al., 2022) offers a more guided workflow but relies on detailed user input at the file level, which becomes impractical for large heterogeneous datasets.

Bidsme (Beliy et al., 2023) is a general-purpose converter that assumes datasets are already organized in a structured hierarchy prior to conversion. While effective when data are consistently arranged, many EEG datasets are stored using vendor-specific naming schemes or lab-driven folder structures that do not conform to any standard hierarchy. In such cases, substantial manual reorganization is required before conversion can begin.

A natural question is whether SOVABIDS' goals could have been achieved by contributing to an existing tool, particularly MNE-BIDS. We argue they could not. MNE-BIDS is designed around a scripting paradigm where conversion logic is expressed in Python code. Adding rule-based, configuration-driven automation as a non-breaking extension would require architectural changes that diverge from MNE-BIDS' current scripting-oriented design philosophy and typical usage patterns. The same barrier applies to FieldTrip and EEGLAB, where conversion is tightly coupled to their respective analysis environments.

SOVABIDS instead introduces a distinct approach: an explicit two-tier separation between dataset-level rules and participant-level mappings, encoded in human-readable YAML configuration files. Like other tools, SOVABIDS requires dataset-specific configuration, but this configuration is defined once at the dataset level and then automatically propagated to generate participant-level mappings, eliminating the need for per-file scripting or manual intervention. Combined with a flexible, semi-automatic API for extracting BIDS entities from arbitrary file paths (via paired examples, placeholder templates, or regular expressions), this design allows SOVABIDS to handle the heterogeneous directory structures and vendor-specific naming conventions common in real-world EEG datasets. While the YAML-based approach introduces its own modest learning requirement, it offers a declarative alternative to general-purpose scripting that is inspectable, shareable, and auditable, preserving full provenance for reproducible conversion workflows.

Software Design

Developing an EEG-to-BIDS conversion tool requires balancing usability, automation, reproducibility, and flexibility while ensuring compatibility with existing neuroimaging tools. The central design tension in SOVABIDS is between expressiveness and accessibility: more powerful conversion logic typically requires programming skill, while simpler interfaces tend to sacrifice flexibility. The following five design principles reflect the trade-offs made to resolve this tension.

1. Accessibility for non-technical users

A scripting-based interface, as used by tools like MNE-BIDS, offers maximum expressiveness but requires users to write and maintain dataset-specific code. We traded this expressiveness for accessibility by using human-readable YAML configuration files instead. This approach was inspired by Bidscoin (Zwiers et al., 2022), a BIDS converter for MRI data. While YAML introduces its own learning requirement for users unfamiliar with the format, it offers a declarative configuration that is reusable across similar datasets and auditable without programming knowledge. The trade-off is that highly unusual conversion scenarios may require more verbose configuration, but for the vast majority of EEG datasets the YAML-based

approach is sufficient and considerably more approachable. To further lower the barrier to adoption, step-by-step guides and usage examples are provided in the documentation.

2. Automation that can accommodate outliers

EEG experiments typically produce multiple identically-organised datasets, one per participant. In practice, however, data organisation often varies slightly between participants due to technical issues, partial recordings, or repeated segments. A fully automated system that assumes identical structure across participants would silently fail in these cases, while a fully manual system would not scale. SOVABIDS resolves this by separating conversion logic into two configuration files (illustrated in Figure 2):

- The **Rules File**, which encodes general conversion rules for the full dataset.
- The **Mappings File**, populated from the Rules File, which holds specific conversion parameters for every individual file.

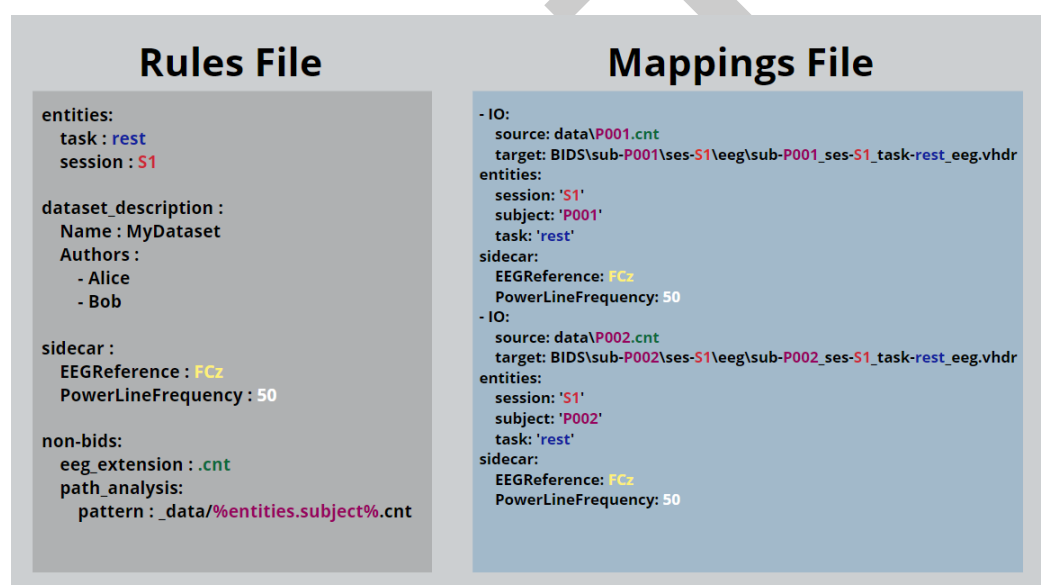


Figure 2: From a Rules File, a mapping for each file in the dataset can be generated and saved in the Mappings File. The colors illustrate how the information in both files is related.

This two-tier approach, inspired by tools such as Bidscoin (Zwiers et al., 2022) and HeuDiConv (Halchenko et al., 2023) (both focused on MRI), extends their model by explicitly supporting non-identical participant structures. Users can also derive an initial Rules File from a community or institutional template to further reduce manual input, and can connect an external GUI via SOVABIDS' API for supervised adjustment of edge cases where full automation is not possible.

To support flexible metadata extraction, SOVABIDS provides a semi-automatic API for inferring subject, session, task, and other BIDS-relevant properties from arbitrary file paths. Rather than requiring data to be pre-organized into a standard folder hierarchy before conversion can begin, as both Bidscoin and Bidsme do, SOVABIDS extracts these properties through pattern matching that is defined once at the dataset level and applied automatically across all files. This is supported through three approaches of increasing technicality: paired source-target examples for users unfamiliar with pattern matching, placeholder-based templates for intermediate users, and full regular expressions for advanced users who need precise control.

3. Reproducible conversion

Reproducibility requires that the full conversion be re-runnable from saved state alone, without relying on the user's memory or undocumented manual steps. All parameters needed to replicate a conversion are therefore saved in the configuration files alongside provenance information. This allows users to audit, correct, and re-run conversions when a BIDS validator flags structural issues or when downstream analysis reveals incorrect metadata.

4. Accessible interfaces and interoperability

SOVABIDS provides two interactive access paths beyond the CLI and Python API, targeting different user needs and deployment environments.

The primary interactive interface is a built-in terminal user interface (TUI), launched via the `sovatui` command. The TUI guides users through the full conversion workflow in four steps — Setup, Rules, Mappings, and Convert — without requiring any code. The TUI was preferred over both web-based and native GUI alternatives for three reasons. First, portability: unlike a web GUI (which requires a running server and a browser) or a native desktop application (which requires platform-specific packaging and distribution for Windows, macOS, and Linux separately), the TUI runs anywhere a terminal is available, including HPC clusters and remote servers accessed over SSH — the environments where large-scale EEG processing most commonly takes place. Second, dependency footprint: native GUI frameworks such as Qt add large binary dependencies and platform-specific installation complexity; the TUI implementation is a single pure-Python file using the `textual` library. Third, maintenance: web and native GUIs accumulate platform-specific bugs, packaging pipelines, and frontend toolchains; a terminal interface alleviates these. A walkthrough is available at <https://youtu.be/dOWiMTuGvAA>.

For external integration, SOVABIDS exposes an RPC-based API that allows any external application — desktop, web-based, or platform-specific — to interact with its conversion logic. RPC was chosen over REST because its action-oriented design maps naturally onto the procedural steps of a data conversion workflow. To demonstrate the API's usability, a [reference web GUI was developed in Flask](#) and is available as a working [example](#). This path is intended for platforms that wish to embed SOVABIDS in a richer desktop or web frontend.

5. Format support through MNE delegation

Supporting the full range of EEG hardware formats from scratch would be an ongoing maintenance burden disproportionate to the tool's core contribution. SOVABIDS instead delegates file reading entirely to MNE-Python ([Gramfort et al., 2013](#)), and BIDS-compliant saving to MNE-BIDS ([Appelhoff et al., 2019](#)), inheriting support for all formats those libraries handle. The trade-off is that SOVABIDS' format coverage is bounded by MNE's, but this is an acceptable constraint given MNE's broad and actively maintained format support. In practice, SOVABIDS has been specifically tested with BrainVision (.vhdr), EDF (.edf), EEGLAB (.set), and FIF (.fif). Read-only formats supported by MNE but not writable via its export API (such as Neuroscan .cnt, BDF, KIT, and CTF) are covered by MNE delegation but are not independently tested in SOVABIDS' continuous integration suite. Basic MEG datatype routing is implemented, but MEG-specific BIDS requirements — empty-room recordings, manufacturer calibration files, and digitization coordinate systems — are not currently exposed through SOVABIDS' rule system and must be handled manually.

Architecture Overview

The five design principles above are reflected directly in SOVABIDS' two-module architecture, illustrated in [Figure 3](#). The Rules Module takes the user-defined Rules File and applies it across all EEG files in the dataset, extracting conversion parameters and compiling them into a Mappings File. This separation means that general conversion logic and participant-specific details are handled in distinct, inspectable artifacts rather than embedded in code. The

179 Conversion Module then reads the Mappings File and performs the actual transformation
180 to BIDS-compliant output, delegating file reading to MNE and BIDS-compliant saving to
181 MNE-BIDS. Users can interact with both modules through the CLI, the Python API, or the
182 built-in TUI. At either stage, the RPC API additionally allows external tools and GUIs to
183 inspect or modify the configuration, supporting the supervised adjustment workflows described
184 above.

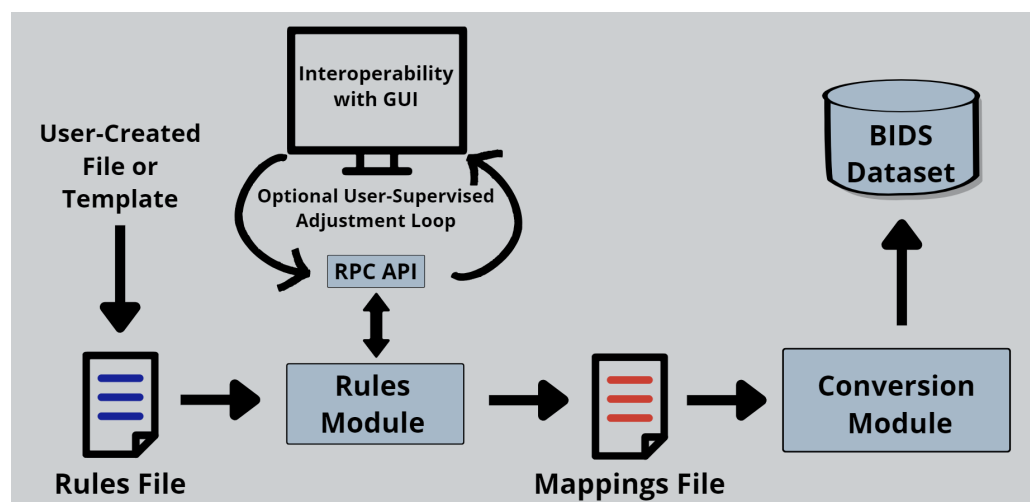


Figure 3: The architecture of SOVABIDS. The conversion process starts with a user-defined Rules File, which encodes general conversion rules (represented in blue inside the Rules File). The Rules Module processes these rules to generate a Mappings File, which contains specific configurations for all EEG files (each red line in the Mappings File represents the configuration of a different file). The Conversion Module then applies these configurations to produce a BIDS-compliant dataset. Interoperability is enabled via an RPC API, allowing integration with external tools, including graphical user interfaces for optional user-supervised adjustments.

185 Research Impact Statement

186 SOVABIDS is listed in the [official BIDS converter registry under the EEG/MEEG/iEEG category](#),
187 reflecting recognition by the broader BIDS community. It is also available on the Neurodesk
188 platform www.neurodesk.org (Dao et al., 2025; Renton et al., 2024), a community-maintained
189 open neuroimaging environment, which broadens its accessibility beyond the developing team.

190 Its use has been documented in peer-reviewed and academic work: a Master's thesis on EEG-
191 based Alzheimer's risk classification (Henao Isaza, 2023), a Bachelor's thesis on web-based
192 EEG processing tools (Zapata Saldarriaga, 2022), and a peer-reviewed study on harmonizing
193 EEG features across multiple recording sites (Jaramillo-Jimenez et al., 2024). These use cases
194 reflect SOVABIDS' applicability across different research scales, from individual thesis projects
195 to multi-site data harmonization studies.

196 Acknowledgements

197 The authors acknowledge the support from the 2021 Google Summer of Code program
198 under the International Neuroinformatics Coordinating Facility (INCF) organization, and the
199 funding provided by the Australian Research Data Commons (ARDC) to support the Australian
200 Electrophysiology Data Analytics Platform (AEDAPT). The authors also acknowledge the
201 facilities and the scientific and technical assistance of the National Imaging Facility, a National
202 Collaborative Research Infrastructure Strategy (NCRIS) capability, at Swinburne Neuroimaging,

203 Swinburne University of Technology, and at the Centre for Advanced Imaging, The University
204 of Queensland.

205 AI Usage Disclosure

206 The primary architecture and core functionality of this software were completed prior to
207 December 2023. Generative AI tools were not used in the conceptual design, methodological
208 decisions, or scientific development of the project.

209 After March 2025, limited use of generative AI tools was made for maintenance and supporting
210 tasks. The tools used were ChatGPT Codex (o4-mini) and ChatGPT (GPT-4.1) and Claude
211 (Sonnet 4.6).

212 AI assistance was used for updating and improving GitHub Actions continuous integration
213 workflows, enhancing and clarifying documentation, assisting in the implementation of a
214 utility function for generating random 1/f signals, minor code refactoring and formatting
215 improvements, and minor improvements on the paper.

216 All AI-assisted outputs were carefully reviewed, edited, tested, and validated by the authors.
217 All core design decisions and scientific judgments were made by the human authors.

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